Tech Tangle - Question Bank

TWX39 Tech Tangle | 200 Questions (Short Answer)

Section A: Computer Fundamentals

1. How many bits make up a nibble?

2. Convert the binary number 10110110 to hexadecimal.

3. What is the 8-bit binary representation of decimal 255?

4. Which logic gate outputs 1 only when its two inputs are different?

5. Name the two 'universal' logic gates.

6. What is the 8-bit two's complement representation of -1?

7. Which is the fastest memory inside a computer system?

8. Expand BIOS.

9. Which firmware standard has largely replaced BIOS on modern PCs?

10. Which CPU component performs arithmetic and logical operations?

11. Which CPU register holds the address of the next instruction to execute?

12. What is the name of the small, fast memory placed between CPU and RAM to store frequently used data?

13. Which generation of computers is associated with integrated circuits (ICs)?

14. Who is called the 'Father of the Computer' for designing the Analytical Engine?

15. Which scientist is credited with the stored-program computer architecture?

16. Which architecture uses physically separate memory for instructions and data?

17. What does RAID level 1 do with data?

18. What is the minimum number of disks needed for RAID 5?

19. What is virtual memory?

20. What is the term for severe slowdown caused by excessive paging between RAM and disk?

21. Which CPU scheduling algorithm can cause starvation of long processes?

22. Name the four necessary conditions for deadlock.

23. NAND flash memory is erased in units of what name?

24. What is the maximum disk size supported by the MBR partition scheme?

25. Which file system is the default for modern Windows installations?

26. Which file system is the common default on many Linux distributions?

27. How many bytes are there in 1 KiB?

28. Which Unicode encoding is backward compatible with ASCII?

29. What is the decimal ASCII value of the character 'A'?

30. Standard (original) ASCII uses how many bits per character?

31. What is the kernel of an operating system?

32. Which Linux command displays running processes dynamically in real time?

33. What is a 'zombie process'?

34. Which Unix system call creates a new process?

35. Convert binary 111101 to octal.

Section B: Programming Languages

36. Python: What is the output of print(type([]))?

37. Python: What is the output of print(3 * '2')?

38. Python: What is the output of print(2**3**2)?

39. Python: What does print(0.1 + 0.2 == 0.3) display?

40. Python: What is the output of print(bool('False'))?

41. Python: What does len({1, 2, 2, 3}) return?

42. Python: What does print(10 // 3, -10 // 3) output?

43. Python: What is the value of [1, 2, 3][-1]?

44. Python: Which keyword creates an anonymous function?

45. Python: Is a tuple mutable or immutable?

46. Python: What does the 'is' operator compare?

47. Python: What is the output of print(type(None))?

48. Python: Which symbol is used to apply a decorator?

49. Python: A function that uses 'yield' returns what kind of object?

50. Python: What does print('a' 'b') output?

51. C: What is the typical size of an int on a modern 32/64-bit system?

52. C: What does the declaration 'int *p;' mean?

53. C: What does printf("%d", 5/2); print?

54. C: Which function releases dynamically allocated memory?

55. C: What is a dangling pointer?

56. C: Which header file declares printf()?

57. C: What is the difference between ++i and i++?

58. Java: Who is considered the creator of Java?

59. Java: Is Java pass-by-value or pass-by-reference?

60. Java: Which keyword prevents a class from being inherited?

61. Java: Expand JVM.

62. Java: What is the default value of an int instance variable?

63. Java: Which class is the parent of all classes?

64. JavaScript: What does typeof null return?

65. JavaScript: What is the result of [] + []?

66. JavaScript: What does 0 == '0' evaluate to?

67. JavaScript: What does NaN === NaN return?

68. JavaScript: Name the two keywords for block-scoped variables.

69. Which runtime lets JavaScript run outside the browser on servers?

70. SQL: Which clause filters groups after GROUP BY?

71. SQL: Which command removes a table's structure and data entirely?

72. SQL: How does COUNT(column) differ from COUNT(*)?

73. SQL: What does NULL = NULL evaluate to?

74. What is the time complexity of binary search?

75. Which data structure follows LIFO order?

Section C: Software Applications and Tools

76. MS Excel: Which function looks up a value in the first column and returns a value from the same row?

77. MS Excel: Which newer function is the modern replacement for VLOOKUP/HLOOKUP?

78. MS Excel: What does COUNTA count?

79. MS Excel: Which symbol makes a cell reference absolute?

80. MS Excel: Which error appears when you divide by zero?

81. MS Excel: Which error appears when VLOOKUP finds no match?

82. MS Word: Which feature creates personalised letters from a list of recipients?

83. What is the default file extension of a Word 2007+ document?

84. What is the default extension of a PowerPoint 2007+ presentation?

85. Which Excel extension indicates a workbook that can contain macros?

86. What does Ctrl + Shift + Esc open in Windows?

87. Which Windows shortcut locks the computer?

88. Is Adobe Photoshop primarily raster-based or vector-based?

89. Which type of graphics is Adobe Illustrator designed for?

90. Which image format supports lossless compression with transparency?

91. JPEG uses which type of compression?

92. Expand SVG.

93. Which Git command copies a remote repository to your machine?

94. What is the difference between git fetch and git pull?

95. Which company owns GitHub?

96. Which free, open-source office suite is an alternative to MS Office?

97. What is SaaS? Give one example.

98. Expand API.

99. Expand PDF and name the company that created it.

100. Which package manager is used on Debian/Ubuntu Linux?

101. Which part of the host do Docker containers share, unlike virtual machines?

102. What file typically defines the instructions for building a Docker image?

103. What does 'CRUD' stand for in database applications?

104. Which markup language is used to structure web pages?

105. Which language is used to style web pages?

Section D: Networking

106. How many layers does the OSI model have?

107. At which OSI layer does a router operate?

108. At which OSI layer does a switch primarily operate?

109. Which transport protocol is connection-oriented: TCP or UDP?

110. What is the default port number for HTTPS?

111. What is the default port for SSH?

112. What is the default port for DNS?

113. What is the default port for SMTP?

114. What is the default control port for FTP?

115. What does HTTP status code 404 mean?

116. What does HTTP status code 301 mean?

117. What does HTTP status code 500 mean?

118. Which HTTP status code means authentication is required: 401 or 403?

119. How many bits are in an IPv4 address?

120. How many bits are in an IPv6 address?

121. How many usable host addresses does a /24 network have?

122. How many usable host addresses does a /30 network have?

123. Which private IPv4 block is commonly used in home routers?

124. What is the 127.0.0.1 address called?

125. A Windows PC shows an address starting with 169.254. What does that indicate?

126. Which protocol maps an IP address to a MAC address?

127. Which protocol automatically assigns IP addresses to devices?

128. Which command traces the route to a host in Windows?

129. Which Wi-Fi marketing name corresponds to IEEE 802.11ax?

130. How many bits long is a MAC address?

131. What is the standard Ethernet MTU?

132. What does NAT do?

133. Name the three steps of the TCP handshake.

134. Which protocol does the ping command use?

135. Which network topology uses a central hub/switch?

136. Who invented the World Wide Web?

137. Which DNS record maps a domain name to an IPv4 address?

138. Which DNS record specifies a domain's mail servers?

139. Expand CDN.

140. What is the difference between bandwidth and latency?

Section E: Cybersecurity

141. What does the CIA triad stand for in security?

142. What is phishing?

143. What is vishing?

144. What is smishing?

145. What does ransomware do?

146. What is a DDoS attack?

147. Which coding technique best prevents SQL injection?

148. Expand XSS.

149. Expand CSRF.

150. Name the three types of authentication factors.

151. Which is reversible with a key: hashing or encryption?

152. Is AES symmetric or asymmetric?

153. Is RSA symmetric or asymmetric? What hard problem does it rely on?

154. Which protocol provides the encryption in HTTPS?

155. What is 'salting' in password storage?

156. Why is MD5 considered unsafe for security use?

157. What is the output length of SHA-256?

158. What is a zero-day vulnerability?

159. What is social engineering?

160. What is a man-in-the-middle attack?

161. What is a honeypot?

162. What does a firewall do?

163. What is the difference between IDS and IPS?

164. What is the purpose of a VPN?

165. What is the principle of least privilege?

166. What is a botnet?

167. What does a keylogger do?

168. Which popular tool is used for network scanning and port discovery?

169. Which popular tool captures and analyses network packets?

170. What is the OWASP Top 10?

171. Expand CVE.

172. What is a rainbow table?

173. What is Stuxnet?

174. Which attack uses '../' sequences to access files outside the intended folder?

175. WannaCry ransomware spread using which exploit/protocol weakness?

Section F: Current Technology Trends

176. Expand LLM in AI.

177. Expand GPT.

178. Which neural architecture was introduced in the 2017 paper 'Attention Is All You Need'?

179. Expand RAG in the context of AI.

180. What is an AI 'hallucination'?

181. Which company develops the Claude AI models?

182. Which company develops the Gemini AI models?

183. Which company makes the H100 GPUs widely used for AI training?

184. What is the basic unit of information in quantum computing?

185. Expand IoT.

186. What is edge computing?

187. What is a smart contract?

188. What is the pseudonym of Bitcoin's creator?

189. Which consensus mechanism did Ethereum move to in 2022 ('The Merge')?

190. Expand NFT.

191. Which spatial-computing headset did Apple release in 2024 in the US?

192. Expand GDPR and name the region it applies to.

193. Which Indian law enacted in 2023 governs digital personal data?

194. Expand UPI.

195. Which organisation developed UPI?

196. Chandrayaan-3 made a soft landing in which year, near which region of the Moon?

197. What are 'deepfakes'?

198. What does 'serverless' computing mean?

199. What is a digital twin?

200. Which IEEE standard number corresponds to Wi-Fi 7?